Sang-yun Lee (Yun)

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EDUCATION

Master of Integrated Product Design (M:IPD) | University of Pennsylvania Expected May 2028
- Relevant Coursework: Product Engineering, Human Systems Engineering, Accessibility Design, Rehab Engineering & Design, Ergonomics

Master of Service Design (GPA: 4.3/4.5) | Sungkyunkwan University Sep. 2022 – Aug. 2025
- Relevant Coursework: Product-Service Convergence Design, Advanced Convergence Service-Data Design, Product Anatomy and Consumer Studies, Social Service Innovation Design, Service Design Processes, Understanding Children with Disabilities
- Thesis: *Data-driven Disability Persona Augmentation for Service Journey Inference*: Daily Living Support Needs of People with IDD

Dual Degree: B.B.A. & B.S. in Data Science (GPA: 3.95/4.5) | Sungkyunkwan University Mar. 2017 – Feb. 2022
- Relevant Coursework: Product/Service Development for Startups, Data Marketing, Platform Revolution and Business, Financing in Cultural Industry, Practice in Data Visualization, User Experience Capstone Design, Location-based Data Analytics, Intro to NLP, Applied Machine Learning, Strategic Management

TECHNICAL SKILLS

Research: Qualitative & Quantitative Synthesis, Interviews, Field Studies, Usability Testing, Surveys, Thematic Analysis, Behavioral Data Analysis, Persona Development, Segmentation/Clustering, Network Analysis, Process Mining, Journey Mapping, Co-Creation, A/B Testing

Design & Strategy: Service Design, Product Strategy, Market Research, Competitive Analysis, Scenario Planning, Interaction Design, Accessibility, Service Blueprinting, Data Visualization, Wireframes, Rapid Prototyping, 3D Printing, Laser Cutting, Digital Prototyping

Software & Analytics: Excel, Power BI, Tableau, Python, R, SPSS, Figma, Miro, Github, Unity, Magma, Rhino, Onshape, SolidWorks, Adobe Creative Suite, Vibe Coding, Generative Agents, Simulation

Languages: Korean (Native), English (Fluent), Chinese (Conversational)

PROFESSIONAL EXPERIENCE

Business Developer, Market Intelligence Team (Customer Survey + Market Demand Analysis) **Jul. 2025 – Jul. 2026**
Samsung Electronics | Digital Appliances Division *Suwon, Korea*
- Analyzed global appliance market trends (market size, pricing, market share, B2B/B2C, etc.) across 12 regions and 8 product categories using GfK, Euromonitor, and BSRIA data, and produced 5-year demand forecasts through 2030, informing company-wide/regional strategy.
- Built an enterprise demand-intelligence dashboard and AI assistant, consolidating market, regional, and sales data into on-demand analyses and executive summaries for product management and sales teams; now deployed company-wide and in active use.
- Identified an opportunity for hands-free cooking support and conceived an AI-powered refrigerator service that saves recipes from personal collections & guides users through cooking via voice. Defined the concept, scope, and business model. Directed software engineers to execute.

Data-driven UX Researcher & Service/Experience Designer **Feb. 2021 – Jul. 2025**
SDI Lab | Data-Driven Service Design Business Strategy Consulting Institute (SKKU) *Suwon, Korea*
- Conducted qualitative research including home-visit interviews, co-creation workshops, and persona interviews with people with disabilities, hospital patients, ministry staff, and mall visitors to uncover user needs.
- Analyzed behavioral data from social media, card transactions, and online communities using network analysis, clustering, process mining, and heatmaps to derive customer personas and journey patterns.
- Translated research findings into journey maps, service blueprints, and strategies for clients in retail, healthcare, and the public sector.
- Built generative-agent simulations of persona clusters to automate service-needs assessment for people with developmental disabilities.

First Lieutenant, Headquarters Battery Commander **Mar. 2022 – Jun. 2024**
ROKA | Republic of Korea Army *Injae, Korea*
- Led 60+ soldiers across six functions (transportation/ammunition/communications/medical/operations/intelligence). Built a 3D terrain model to train soldiers on enemy infiltration routes, defending the battalion command post in a war-game against a tank assault and 15 raids.
- President of Korea ROTC 60th Officer Corps (3,561 alumni) — arranged 10 hospital partnerships & a job fair with Shinhan Financial Group.

Research Analyst **Oct. 2021 – Dec. 2021**
BCG | Boston Consulting Group *Seoul, Korea*
- Analyzed manufacturing process, core-component supply chain, and regional regulations (e.g., ISO) to assess market-entry feasibility.
- Developed market-entry strategy scenarios based on the market analysis, including M&A target recommendations.

PROJECTS

SDI Lab | Data-Driven Service Design Business Strategy Consulting Institute (SKKU) *Suwon, Korea*

Aging with Disability Product/Service Design, National Rehabilitation Center **Aug. 2024 – Dec. 2024**
- Co-designed and conducted home-visit interviews with 9 people with spinal cord injury, developmental disabilities, or stroke.
- Synthesized findings into daily-living journey maps for each group that surfaced unmet needs and aging-related challenges.

- Directed a 40-student design hackathon to develop real products/services based on research; guided students to use the journey maps to identify opportunity areas and prototype 9 rehabilitation product/service concepts, which the center evaluated and awarded.

VIP Concierge Robot Service Design, SKKU President's Office **Oct. 2024 – Jun. 2025**
- Owned experience design for VIP reception robot service, interviewing secretarial staff, the president, and VIP visitors to translate their wants into an end-to-end experience blending university's royal-court heritage with its technology.
- Designed 3 coordinated robot personas for welcome, tour, and refreshment, grounded in royal court protocol and the office's existing greeting workflow; visualized them in service blueprints and directed mechanical engineering students on the build; the robots now operate.

Inpatient Rehabilitation Service Design, Samsung Hospital (SAIHST) **Aug. 2024 – Dec. 2024**
- Co-designed gamified rehabilitation programs (step-count levels, meal-pickup and exercise missions) and the UI/UX of a hospital app to promote patient rehabilitation, grounded in interviews and surveys with long-term hospitalized patients.
- Conceived and prototyped in Unity and Magma a collaborative artwork service where patients find tags around the hospital like a treasure hunt, collect puzzle pieces, and combine them into a shared artwork displayed in public hospital spaces.

Mega Shopping Mall Service Design, Shinsegae Starfield Shopping Mall **Feb. 2021 – Dec. 2021**
- Conducted mixed-method research to define brand personality per branch via NLP-based social data analysis and within-branch customer personas via behavioral data analysis (demographic, network, process mining) with Power BI and persona interviews.
- Proposed mall-level service strategies based on the derived personas and journeys, including F&B-centered customer journeys, store flow and tenant changes, and social event planning, adopted in new branch operations.

Mini-Fridge Subscription Service Design, Samsung Electronics **Mar. 2023 – Jun. 2023**
- Classified users by lifestyle to devise mini-fridge sales strategies, analyzing life-log data from diary studies and focus group interviews through process-mining journey maps, network analysis, and heatmaps of living-radius patterns to derive 7 personas.
- Proposed a lifestyle-based subscription service that stocks mini-fridges with products matched to each derived persona.

Digital Public Communication Strategy Design, Ministry of Culture, Sports & Tourism **Jul. 2021 – Dec. 2021**
- Designed citizen-participatory policy communication to address the ministry's ineffective social media information delivery.
- Led a 3-day co-creation workshop (ministry staff + students) via Miro-based card sorting & network-based context visualization to reveal how citizens interpret policy messaging; co-designed improved content; selected as an OECD Best Practice in public policy communication design.

AIEX Design Talent Program, SKKU (National R&D Project) **Jul. 2024 – Jun. 2025**
- Organized a 118-student design-talent training program — hackathons, startup design-thinking school, Tokyo capstone.

PUBLICATIONS

Data-driven Disability Persona Augmentation for Service Journey Inference
Published, Sungkyunkwan University Jun. 2025
- Developed a computational framework for studying disability service needs where experiments and data are hard to obtain.
- Augmented Vineland-adaptive behavior data from 9 individuals with developmental disabilities to 2,400 cases; derived 5 persona clusters and ran AI-driven daily life scenario (ADL/IADL) simulations that compute support-need levels and causes, visualized as a journey map.

Patterning Military Experience for Quality Improvement of Enlisted Soldiers
Published, Korean Academy of Military Social Welfare Dec. 2024
- Developed a pattern model of high-performing soldiers' military life to give soldiers who struggle to adapt a clear path to follow.
- Interviewed 18 high-performing soldiers, combining longitudinal direct observation; yielded a 5-stage military life-cycle pattern model.